PROBLEM 32

# Particle with String on Spool

Original statement · Problems.pdf, p. 20

Rheonomic Spool Constraint

A heavy particle $P$ of mass $m$ is connected by a string of total length $l = 4\pi a$ to a cylindrical spool of radius $a$ and center at $O$. The spool rotates at a known constant angular rate $\omega$ counterclockwise. The string is massless, inelastic, thin, and remains always in tension during the motion of the particle. We define the inertial reference frame $Oxy$. The other end of the string is attached to the spool at a point $A$ of its perimeter, and we shall call $B$ the instantaneous point of tangency between the straight segment of the string (the segment $BP$) and the spool. We will denote the angle between the local horizontal (axis $Ox$) and the radius $OB$ as $\phi$. Initially, $\mathbf{OA} = -a\mathbf{i}$, and $B$ never reaches $A$.

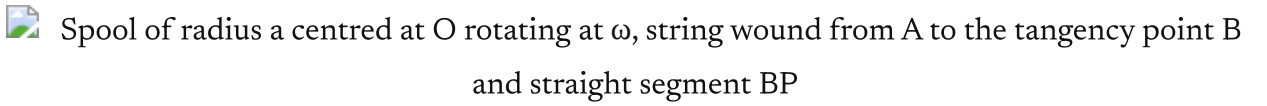



*Figure: Spool of radius a centred at O rotating at ω, string wound from A to the tangency point B and straight segment BP (Problems.pdf, p. 20).*

**(a)** Discuss the number of degrees of freedom of the motion of the particle.

**(b)** Write down the position, velocity, and acceleration vectors of $P$ with respect to $Oxy$ as a function of the constants of the problem, time, the angle $\phi$, and its derivatives.

**(c)** Write down the mathematical expression of the forces acting on $P$.

**(d)** Is the mechanical energy of $P$ conserved? If so, write down the equation of conservation of mechanical energy; otherwise, explain why it is not conserved.

**(e)**

Write down a closed set of equations that, if solved, would yield $\phi$ (and the other unknowns of the problem, if any) as a function of time. Note that you do not need to solve the equations.

**(f)** Consider now that $\omega = 0$. Can you write down a first integral of motion that only involves $\phi$ and $\dot{\phi}$ (but not $\ddot{\phi}$) in this case? You may want to rethink your previous answers for this particular case.

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**Full Analytical Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle $P$ of mass $m$ is attached to a string of length $l = 4\pi a$ unwinding from a circular spool of radius $a$ rotating counterclockwise at constant rate $\omega$.
- **Initial Tangency:** Initially at $t = 0$, $A$ is at $(-a, 0)$ and the string is wrapped around the spool.
- **CDOFs:** 1 degree of freedom: the instantaneous unwinding angle $\phi(t)$.
- **Parameters:** $m, a, l = 4\pi a, \omega, g > 0$.

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- Inertial frame: $Oxy$.
- Position of contact point $B$: $\mathbf{r}_B = a\cos\phi\,\mathbf{i} + a\sin\phi\,\mathbf{j}$.
- Unit vectors at $B$:
- Radial: $\mathbf{u}_r = \cos\phi\,\mathbf{i} + \sin\phi\,\mathbf{j}$.
- Tangential: $\mathbf{u}_t = -\sin\phi\,\mathbf{i} + \cos\phi\,\mathbf{j}$ (counterclockwise tangent; the straight segment leaves $B$ clockwise, towards the lower right in the figure, i.e. along $-\mathbf{u}_t$).
- Straight segment length:

The initial wrapped length was $l$. As the spool rotates at rate $\omega$, the length of the unwound free segment $BP$ is:

$$s(t) = a(3\pi - \omega t + \phi)$$

- Position of particle $P$:

$$\mathbf{r}_P = \mathbf{r}_B - s(t)\mathbf{u}_t = a\cos\phi\,\mathbf{i} + a\sin\phi\,\mathbf{j} - a(3\pi - \omega t + \phi)(-\sin\phi\,\mathbf{i} + \cos\phi\,\mathbf{j})$$

PHASE 3

### Step-by-Step Mathematical Deduction

**Part (a): Degrees of Freedom**

- Although $P$ moves in the plane (2 coordinates), the constraint of remaining tangent to the unwinding spool leaves exactly **1 degree of freedom** ($\phi$).

**Part (b): Velocity and Acceleration Vectors**

Differentiating $\mathbf{r}_P(t)$ with respect to time:

$$\mathbf{v}_P = \frac{d\mathbf{r}_P}{dt} = a\dot{\phi}\mathbf{u}_t - \dot{s}\mathbf{u}_t + s\dot{\phi}\mathbf{u}_r$$

Since $s(t) = a(3\pi - \omega t + \phi) \implies \dot{s} = a(\dot{\phi} - \omega)$:

$$\mathbf{v}_P = a\omega\,\mathbf{u}_t + s\dot{\phi}\mathbf{u}_r$$

Differentiating once more, with $\dot{\mathbf{u}}_t = -\dot{\phi}\mathbf{u}_r$ and $\dot{\mathbf{u}}_r = \dot{\phi}\mathbf{u}_t$:

$$\mathbf{a}_P = \left[s\ddot{\phi} + a\dot{\phi}^2 - 2a\omega\dot{\phi}\right]\mathbf{u}_r + s\dot{\phi}^2\mathbf{u}_t$$

**Part (c): Forces Acting on $P$**

1. Gravity: $\mathbf{F}_g = -mg\mathbf{j} = -mg(\sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_t)$.

2. Tension: $\mathbf{T} = +T\mathbf{u}_t$ ($T > 0$ directed towards $B$, since $\mathbf{r}_B - \mathbf{r}_P = s\,\mathbf{u}_t$).

> **Erratum:** The first version placed the particle at $\mathbf{r}_B + s\mathbf{u}_t$ (up and to the left of $B$), which contradicts the figure where the segment $BP$ leaves $B$ towards the lower right. Correct: $\mathbf{r}_P = \mathbf{r}_B - s\mathbf{u}_t$, $\mathbf{v}_P = a\omega\,\mathbf{u}_t + s\dot\phi\mathbf{u}_r$, $\mathbf{a}_P = [s\ddot\phi + a\dot\phi^2 - 2a\omega\dot\phi]\mathbf{u}_r + s\dot\phi^2\mathbf{u}_t$ and tension $+T\mathbf{u}_t$; the equations of part (e) and the energy of part (f) were already consistent with this. Numeric check: sympy differentiation of $\mathbf{r}_P$ reproduces $\mathbf{v}_P$ and $\mathbf{a}_P$, and RK4 of (e) with $\omega = 0$ conserves the energy of (f) to $10^{-13}$ with $T > 0$.

**Part (d): Mechanical Energy Discussion**

For $\omega \neq 0$, the constraint is **rheonomic** (explicitly time-dependent). The contact point $B$ moves with velocity $\mathbf{v}_B = a\omega\mathbf{u}_t$, so the string tension does work on $P$. Mechanical energy is **not conserved** for $\omega \neq 0$.

**Part (e): Closed Set of Equations**

Projecting along the tangent $\mathbf{u}_t$ and normal $\mathbf{u}_r$:

$$a\left[(3\pi - \omega t + \phi)\ddot\phi - 2\omega\dot\phi + \dot\phi^2\right] = -g\sin\phi$$

$$T = ma(3\pi - \omega t + \phi)\dot\phi^2 + mg\cos\phi$$

Compare with the official solution key (Problems.pdf, Solution keys section).

**Part (f): Case $\omega = 0$ (Static Spool)**

When $\omega = 0$, the constraint is scleronomic and the string does zero work. Mechanical energy is strictly conserved:

$$E = \frac{1}{2}m[a(3\pi + \phi)]^2\dot\phi^2 + mg\left[a\sin\phi - a(3\pi + \phi)\cos\phi\right] = \text{constant}$$

Compare with the official solution key (Problems.pdf, Solution keys section).

PHASE 4

### Physical Interpretation & Dimensional Verification

- The system behaves as an involute pendulum whose effective string length increases or decreases linearly as it unwinds.

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*Source: Problems.pdf (Page 20)*